Hamza Ahmed Siddiqui

Data Scientist · ML Engineer · AI Solutions

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Work Experience

~Research Assistant - Computer Vision **Apr 2026 – Present**
Eckerlab *Göttingen, Germany*
Project: Video Generalization Dataset

- Benchmarked self-supervised vision backbones (DINOv2, V-JEPA) on multi-modal datasets by building automated evaluation pipelines to quantify model performance and generalization across dynamic domains.
- Integrated Segment Anything Model (SAM 3) into the vision pipeline for object detection.
- Architected multi-modal feature extraction pipelines using DINOv2, V-JEPA, and SAM 3 backbones in PyTorch, optimizing GPU memory and dynamic batching for large-scale video processing.
- Quantified fine-grained object bias in action recognition datasets by engineering a multi-camera synchronized capture system (custom 3-camera hardware setup) to record controlled validation data and analyze class-level confidence spikes.

Project: Enhancing Domain Generalization in Video Action Recognition

- Engineered and trained two-stream video classification architectures (I3D ConvNets) to evaluate multi-modal inputs (RGB and Optical Flow) for mitigating background bias in cross-domain generalization.
- Architected modular **PyTorch training pipelines from scratch** incorporating WandB experiment tracking, early stopping, optical flow visualizations, and dynamic model checkpointing.
- Optimized data loader throughput by serializing multi-modal datasets into HDF5 (.h5) format, significantly cutting I/O bottleneck and maximizing GPU utilization.
- Designed a late-fusion inference strategy combining RGB and flow modalities with hyperparameter optimization, improving uAP from 72.8 to 75.2 on heavily imbalanced dataset.

Teaching Assistant - NLP **Oct 2025 – Present**
GippLab *Göttingen,Germany*
Project: Teaching Assistant for courses:

- Deep Learning for NLP, Programming for Data Science

Senior Data Scientist | Data Scientist | Junior Data Scientist **Jun 2020 – Apr 2025**
Afiniti *Istanbul, Türkiye*
Project: Customer Propensity Score Modeling Platform: *Afiniti uses patented AI technology to pair participants in customer interactions in a way that optimizes enterprise goals like revenue.*

- Engineered production data integrity and automated drift-detection systems in Python, R and SQL, reducing silent pipeline failures and safeguarding model reliability across real-time customer pairing systems.
- Architected production-ready ML and data processing pipelines using Python, SQL, and Apache Spark, optimizing ETL throughput for multi-million record customer datasets across cloud environments.
- Designed feature engineering workflows and led reproducible model evaluation experiments.
- Designed and executed time-cycle **A/B testing** frameworks and causal inference models for enterprise customer-agent pairing, driving a **25% incremental revenue** lift over baseline FIFO routing.
- Enabled **13%** sales uplift in AI-agent pairing, by integrating customer purchase-probability scores into the matching platform to prioritize customers based on predicted likelihood of conversion.
- Improved next-purchase propensity prediction by increasing ROC-AUC from **0.75** to **0.85**, by training LightGBM models on binary sale outcomes and engineering customer, interaction, call-center, and demographic features.

Leadership experience:

- Led the transformation of a loss-making business unit into a **$300,000/month** profit making unit.
- Led a team of 6 data scientists; mentored new hires and presented in stakeholder meetings..
- Promoted thrice within the first 3 years.

Education

MSc in Applied Data Science **Apr 2025 – Present**
Georg-August-Universität Göttingen *Germany*
Thesis: "TBD"
Overall Grade: ***1.2***

BSc Economics **Aug 2016 – May 2020**
Lahore University of Management Sciences (LUMS) *Lahore, Pakistan*
Overall Grade: ***3.91****/4.00 (High Distinction)*

Selected Projects

Berlin Airbnb Price Prediction Project **Jul 2025**
ML model to predict Airbnb listing prices using features such as location, property type, amenities, and historical pricing data.

- Implemented Catboost Regressor and optimized performance using hyperparameter tuning, achieving strong test accuracy of ~85%. [Github]

ResumeReviewer-Agent **Oct 2026**
Built an agentic AI resume analyzer using **LangChain (LCEL)**, Gemini, and Streamlit to automate candidate skill gap analysis against target job descriptions.

- Engineered a document parsing pipeline (PyMuPDF/Unstructured) and leveraged **Pydantic** structured JSON outputs to generate strictly typed, actionable resume rewrite suggestions. [Github]

More projects on my webpage: https://hamzasiddiqui10.github.io/

Technical Skills

- **Programming & Data:** Python, R, Java, Scala, C/C++, SQL, PostgreSQL, MongoDB, Redis, Neo4j, InfluxDB,
- **Backend & Pipelines:** Linux, FastAPI, Kafka, Spark, Airflow, Docker, Kubernetes, CI/CD, GitHub Actions
- **Cloud:** Azure Functions, Azure Blob Storage, Azure VMs, Azure PostgreSQL, Azure Key Vault, AWS S3, Hadoop, Spark, Databricks
- **Methodologies:** Machine Learning, Statistical Modeling, Prompt Engineering, Agentic AI, AI-Driven Code Generation.
- **ML & Analytics:** PyTorch, HuggingFace, LoRA, A/B testing, statistical analysis, Neural Networks, Transformers, NLP, Regressions, Decision Trees, Ensemble Models (Random Forest, XGBoost, LightGBM, CatBoost), Bayesian Methods, PCA, GMMs, t-SNE, Feature Engineering, Hyperparameter Tuning, ML workflow, LLMs, Prompt Engineering, Structured LLM Output, Langchain, Langsmith, PyDantic, Gemini API.

Honors & Awards

- LUMS Dean's Honor List, 8 out of 8 semesters
- LUMS Graduation with High Distinction (Top 1% in university)
- 100% Scholarship in GCE Advanced Level (High School): Based on excellent grades in O Level

Languages

English: Native **German:** Beginner **Urdu:** Native **Sindhi:** Native

Other Interests

History, Football, Tennis, Swimming